Lesson Planning

Complete Lesson Plan Book

Class: ICS · Class 12

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Introduction to Databases: Tables and Records

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Unit 1: Introduction to Databases: Tables and Records

Learning Objectives

• Students will explain what a database is and why organised data beats scattered files.

• Students will identify tables, fields, records and a primary key in a simple example.

• Students will design a small table for a real situation with sensible field types.

Key Concepts

• A database is an organised collection of related data stored so it can be searched and updated easily.

• A table stores data about one kind of thing, for example students or books.

• A field is one column of a table, such as name or roll number; a record is one full row.

• A primary key is a field whose value is unique for every record, so each row can be found exactly.

Teaching Methodology (Step by Step)

1. Compare a pile of paper registers with a computer database and list the problems of paper.
2. Draw a students table on the board with fields: roll number, name, class, marks.
3. Fill three records and ask which field can never repeat; name it the primary key.
4. Show how one question, such as students with marks above 80, is answered from the table.
5. Students design their own table for a school library in pairs.

Classroom Activities

• Table build: groups design a table for a cricket team with at least four fields and a primary key.

• Faulty table hunt: find the problem in a table where two students share the same roll number.

• Question race: who can answer three questions from the class table fastest?

Assessment Questions

1. Define a record and a field in one sentence each.
2. Why can a student's name not be a good primary key?
3. Design a table for a shop's customers with four fields and mark the primary key.

Homework

• Design a database table for your class attendance with five fields and a primary key.

• Write two questions your table can answer and one question it cannot.

Time Allocation

40 minutes